Detection and Estimation

Detection - Converting a waveform to a test statistics

Estimation - Mapping test statistics to a Symbol

Detection is a deterministic process and Estimation is a random process.

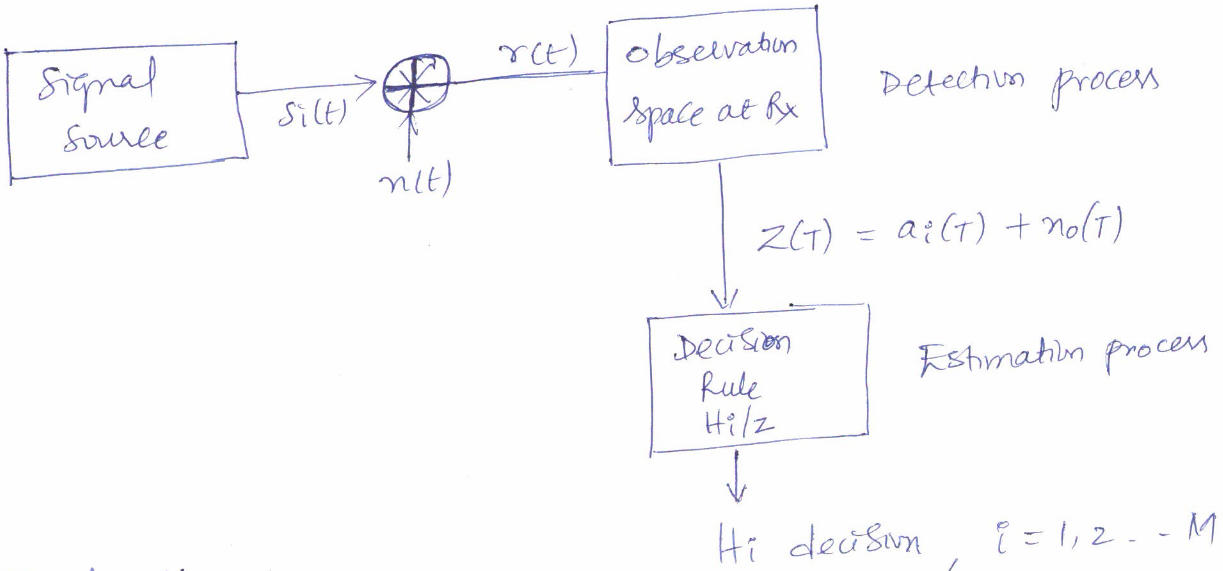
At the Receiver we have

$$r(t) = s_i(t) + n(t)$$

$s_i(t)$ is a deterministic signal with finite Energy and finite bandwidth

$n(t)$ is a random signal with Infinite bandwidth and Infinite energy.

$r(t)$ is a random signal with Infinite bandwidth and Infinite Energy.

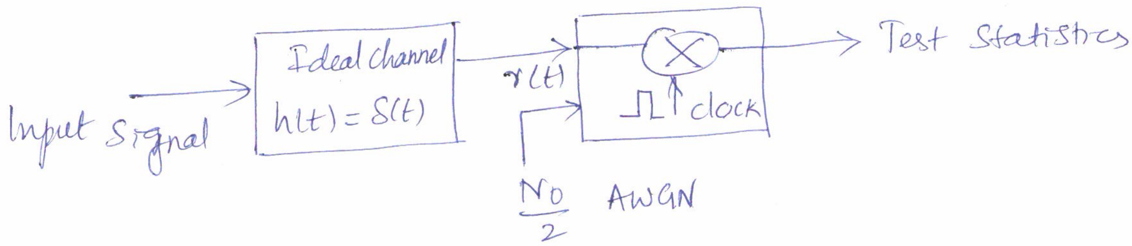


Detection :

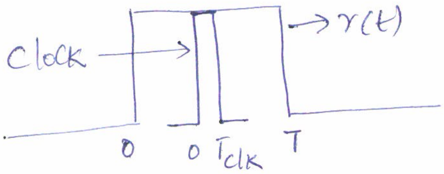
Detection is carried out Using 3-types.

1. Clock Sampling / center Sampling
2. Averaging
3. Matched filter

Clock Sampling

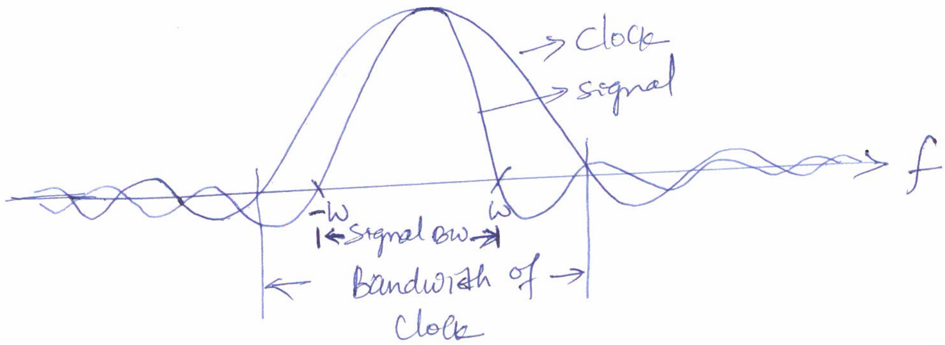


$$r(t) = s_i(t) + n(t)$$



Clock pulse width $T_{clk} << T$ (Symbol duration)

$\therefore$ Clock Bandwidth >> Signal bandwidth

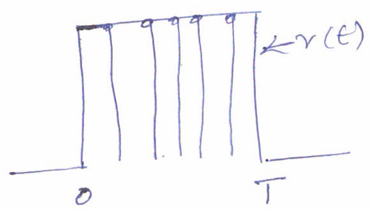


In clock sampling, the noise outside signal range is also coming into picture. It samples only at the center of a pulse.

Test Statistics is noisy in clock sampling.

Averaging – Sample the signal at multiple instants from 0 to T and then average it. Noise also will get averaged.

$r(t) = S_i(t) + n(t)$



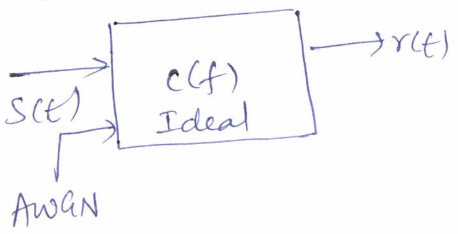
Mean of $n(t) = 0$

Mean of $r(t)$ = Mean of $S_i(t)$

Test statistics is cleaner in averaging as compared to clock sampling. In averaging, we cannot extract the full energy of $r(t)$.

After averaging the signal is $r_i' = \frac{1}{N} \sum_{i=1}^{N} r_i$

When SNR is high and distance of communication is short, for example RS232, 1.5m cable, clock sampling or Averaging is used as a detection purpose.



$|C(f)| = 1$

$R(f) = S(f)\, C(f)$

$r(t) = Re\left[F^{-1}(R(f))\right] = Re\left[F^{-1} |S(f)|\,|C(f)|\, e^{j\, \underline{\angle S(f) + \phi}}\right]$

$r(t) = Re\left[F^{-1} |S(f)|\, e^{j\, \underline{\angle S(f) + \phi}}\right]$

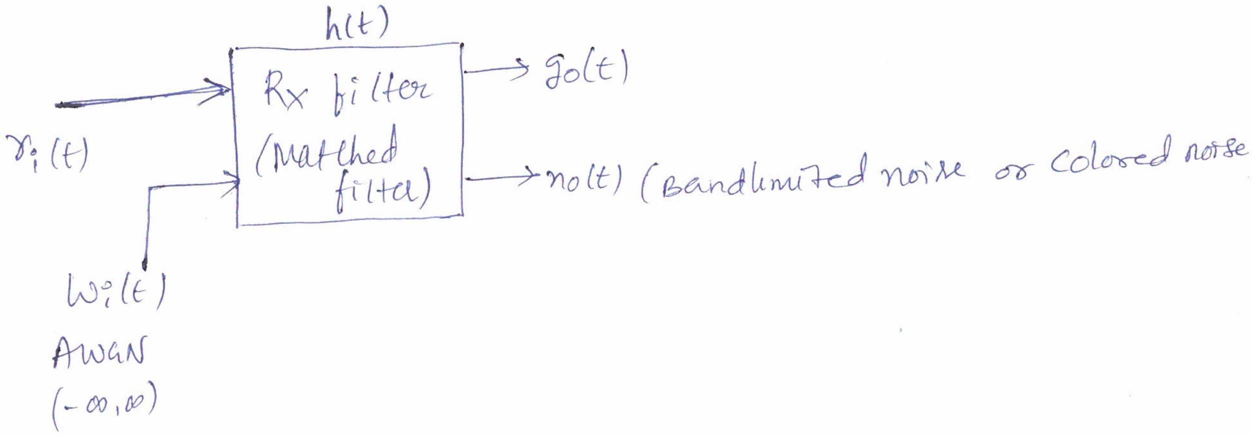
If we can match $\angle\phi = \angle S(f)$, then we can extract the full energy.

When SNR is low and in long distance communication we use matched filter.

Matched Filter — This restrict the noise to signal bandwidth.

It provides a conjugate for the signal, so that we get maximum SNR at the output of the matched filter.

In Matched filter, Test Statistics is cleaner, SNR improves and there will be a better Estimation.



$$r_i(t) = g_i(t) + w_i(t)$$

$$(S/N)_{i/p} = \frac{|g_i(T)|^2}{E[w_i^2(t)]}$$ Noise level is more than the signal level

$$(S/N)_{o/p} = \frac{|g_o(T)|^2}{E[n_o^2(t)]}$$ Signal level will be more than noise level.

Need to establish two conditions

1. What is the maximum SNR possible from a MF
2. What should be the shape of the MF to achieve maximum SNR at the output.

Frequency domain - SNR

$$g_o(t) = g_i(t) * h(t)$$

$$G_o(f) = G_i(f) \cdot H(f)$$

$$g_o(t) = F^{-1}[G_o(f)] = \int_{-\infty}^{\infty} H(f)\, G_i(f)\, e^{j2\pi f t}\, df$$

Matched filter is Sampled at $t = T$

Signal Energy $|g_o(T)|^2 = \left| \int_{-\infty}^{\infty} H(f)\, G_i(f)\, e^{j2\pi f T}\, df \right|^2$

PSD of $n_o(t)$ is $S_N(f) = \frac{N_0}{2} |H(f)|^2$

$$(S/N)_{o/p} = \frac{|g_o(T)|^2}{E[n_o^2(t)]} = \frac{\left| \int_{-\infty}^{\infty} H(f)\, G_i(f)\, e^{j2\pi f T}\, df \right|^2}{\int_{-\infty}^{\infty} S_N(f)\, df}$$

Apply Cauchy - Schwarz inequality - If $\phi_1(x)$ and $\phi_2(x)$ are two complex functions, Integral of products is less than or equal to product of their Integrals. Equality holds when ϕ_1 and ϕ_2 are Conjugate of each other.

$$\int_{-\infty}^{\infty} |\phi_1(x)|^2 dx < \infty, \quad \int_{-\infty}^{\infty} |\phi_2(x)|^2 dx < \infty, \text{ then}$$

$$\left| \int_{-\infty}^{\infty} \phi_1(x)\, \phi_2(x)\, dx \right|^2 \le \int_{-\infty}^{\infty} |\phi_1(x)|^2 dx \cdot \int_{-\infty}^{\infty} |\phi_2(x)|^2 dx$$

Equality holds when $\phi_1(x) = \text{Constant} \cdot \phi_2^*(x)$

Let $\phi_1(x) = H(f)$ and $\phi_2(x) = G_i(f)\, e^{j2\pi f T}$

$$SNR \leq \frac{\int_{-\infty}^{\infty} |H(f)|^2 df \cdot \int_{-\infty}^{\infty} |G_i(f)|^2 df}{\frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 df}$$

$$\leq \frac{2}{N_0} \int_{-\infty}^{\infty} |G_i(f)|^2 df$$

$$SNR \leq \frac{2}{N_0} E_{symbol}$$

Equality holds when $H_{opt}(f) = k \cdot G_i^*(f)\, e^{-j2\pi f T}$

$H_{opt}(f)$ is optimum filter response to maximize the SNR

$$SNR = \frac{2 E_{symbol}}{N_0} \longrightarrow \text{Maximum possible SNR from MF}$$

Impulse response

$$h_{opt}(t) = \mathcal{F}^{-1}\{H_{opt}(f)\}$$

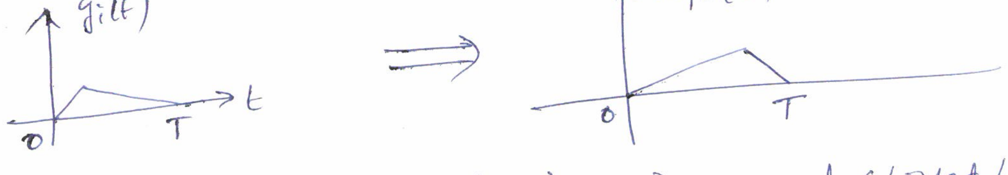
$$= k \int_{-\infty}^{\infty} G_i^*(f) \cdot e^{-j2\pi f T} \cdot e^{j2\pi f t} df$$

$$= k \int_{-\infty}^{\infty} G_i^*(f) \exp(-j2\pi f(T-t))\, df$$

for a real signal $G_i^*(f) = G_i(-f)$

$$h_{opt}(t) = k \int_{-\infty}^{\infty} G_i(-f) \exp(-j2\pi f(T-t))\, df$$

$$= k\, g_i(T-t)$$



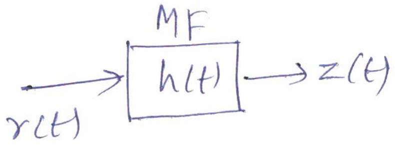
$g_i(t)$ Mirror image and shifted by T

Correlator realization of a Matched filter

Limitations of matched filter are

1. If we have M-Symbols, then we need M-matched filters.
2. MF operation is a Convolution operation.
3. MF output is continuous and available from 0 to 2T.

output of a MF = $z(t) = r(t) * h(t)$



$$r(t) = s(t) + w(t)$$

$$z(t) = \int_0^t r(\tau)\, h(t-\tau)\, d\tau \qquad \text{Convolution}$$

$$h(t) = k\, s(T-t)$$

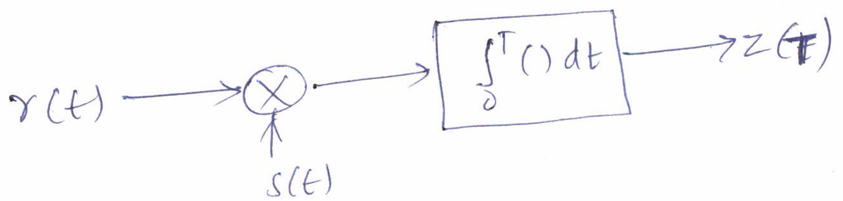
Replace t by $t-\tau$, $h(t-\tau) = k\, s(T-(t-\tau))$

$$z(t) = \int_0^t r(\tau)\, s(T-(t-\tau))\, d\tau$$

$$z(T) = \int_0^T r(\tau)\, s(T-(T-\tau))\, d\tau$$

Matched filter o/p is maximum at $t=T$

$$z(T) = \int_0^T r(\tau)\, s(\tau)\, d\tau \qquad \text{Correlation}$$



It leads to Simple MAC Instruction.

The output of a Correlator is at one shot. Also output is maximum at $t=T$

Properties of a Matched Filter

1. The spectrum of the output of a MF is except the time delay factor is proportional to the energy spectral density of the input signal.

$$S_o(f) = H_{opt}(f)\, S(f)$$

$$S_o(f) = |S(f)|^2 e^{-j2\pi f T}$$

2. The output signal from a MF is proportional to a signal which is shifted version of Autocorrelation function of the input signal.

$$F^{-1}\{S_o(f)\} = F^{-1}\{|S(f)|^2 e^{-j2\pi f T}\}$$

$$s_o(t) = R_s(T-t)$$

3. The o/p SNR of a MF depends on the ratio of signal energy to the PSD of input white noise

$$(S/N)_{o/p} = \frac{2E}{N_0}$$

SNR depends only on signal energy. It doesnot depend on the waveform shape.

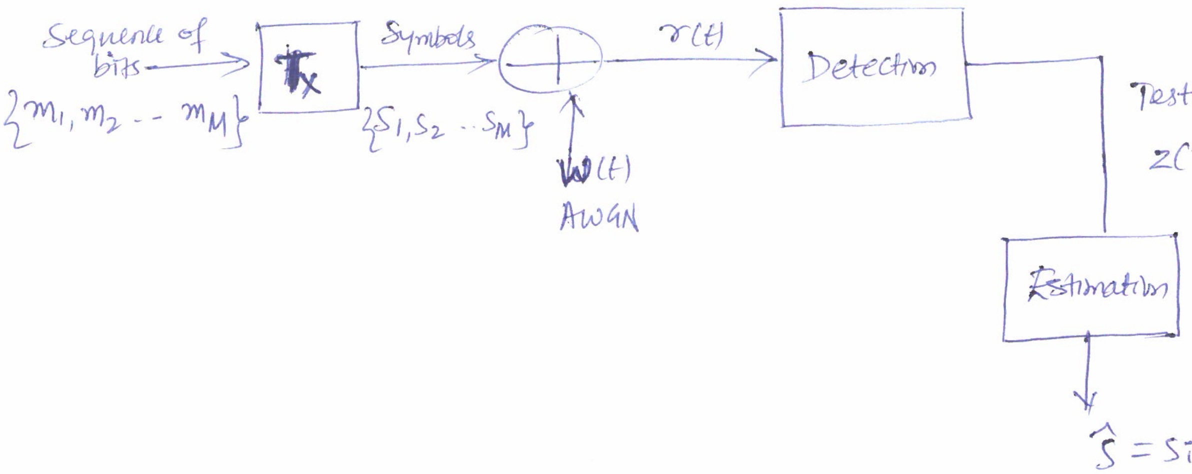
Waveform shape is important when extracting clock for synchronization and finding the bandwidth.

4. MF may be separated into two matching conditions

Spectral amplitude matching $\quad |H(f)| = |S(f)|$

Spectral phase matching $\quad \underline{|H_{opt}(f)} = -\theta(f) - 2\pi f T$

Estimation – process of mapping test statistics to a symbol.



Test Statistics

$z(T) = a_i(t) + n(t)$

Signal sample — Noise sample

$i = 1, 2 \ldots M$

(MAP/MLI)

Two statistical events are observed.

1. Apriori probability $p(s_i)$ at Tx
2. Aposteriori probability $p(s_i/z)$

$$p(s_i/z) = \frac{p(z/s_i)\, p(s_i)}{p(z)}$$ Bayes' theorem

$p(z) = \sum p(z/s_i)\, p(s_i)$ is a common factor, can be dropped

$$p(s_i/z) = p(s_i)\, p(z/s_i)$$

$p(s_i) = \frac{1}{M}$ Equiprobable case

MLI $p(z/s_i) = p(s_i/z)$ MAP

Either maximize MLI or MAP will give lower Pe.

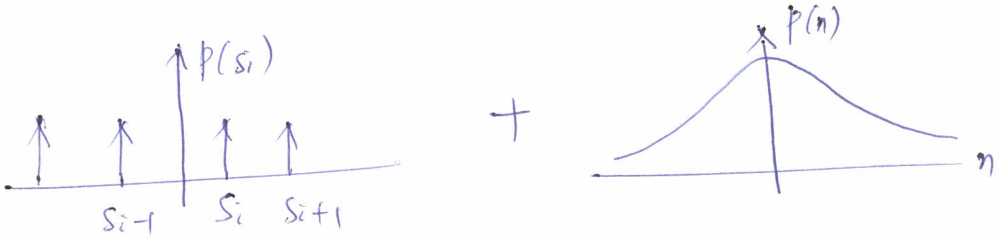
MLI is easier to compute in practice.

Likelyhood Probability $p(z/s_i)$

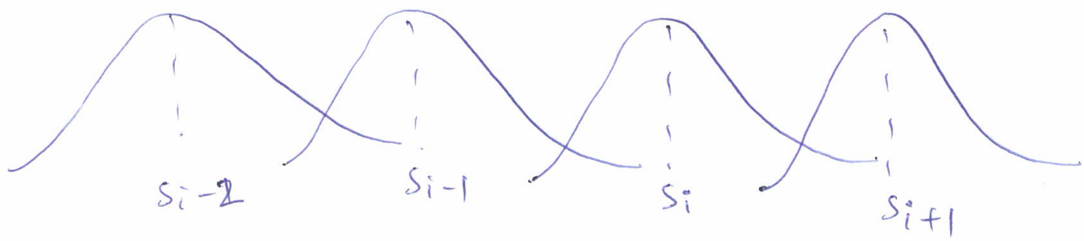
$$z = s_i + n$$

$s_i \rightarrow$ Uniform PDF

$n \rightarrow$ Gaussian PDF



$=$ PDF of z



$$\text{Mean}[z] = E[s_i + n] = E[s_i] = \text{Signal mean}$$

$$\text{Var}[z] = E[(z-\bar{z})^2] = E[(s_i + n - s_i)^2] = E[n^2] = \sigma^2 = \frac{N_0}{2}$$

$\text{Var}[z] =$ variance of noise.

Error probability for $s_i = P_{es_i}$

$$P_{es_i} = p(s_i \text{ sent and } z \neq s_i) = p(s_i)\, p(z \neq s_i / s_i)$$

Two Independent events

$$P_{es_i} = p(s_i)\, p(z \neq s_i / s_i) = p(s_i)\left(1 - p(z = s_i)/s_i\right)$$

But $\quad p(z = s_i / s_i)\, p(s_i) = p(s_i/z)\, p(z) \quad$ Baye's Theorem

$$P_{es_i} = \left[p(s_i) - p(z)\, p(s_i/z)\right]$$

Constant $\quad$ Aposteriori

$$\text{Total } P_e = \sum_{i=1}^{M} P_{e s_i} = \sum_{i=1}^{M} p(s_i)\, p\left(z \neq s_i / s_i\right)$$

$$= \sum_{i=1}^{M} p(s_i)\left[1 - p\left(z = s_i / s_i\right)\right]$$

$$= \sum_{i=1}^{M} p(s_i) - \sum_{i=1}^{M} p(s_i)\, p\left(z = s_i / s_i\right)$$

$$= 1 - \sum_{i=1}^{M} p(s_i)\, p\left(z = s_i / s_i\right)$$

CDF over all possible $z \in S_i$

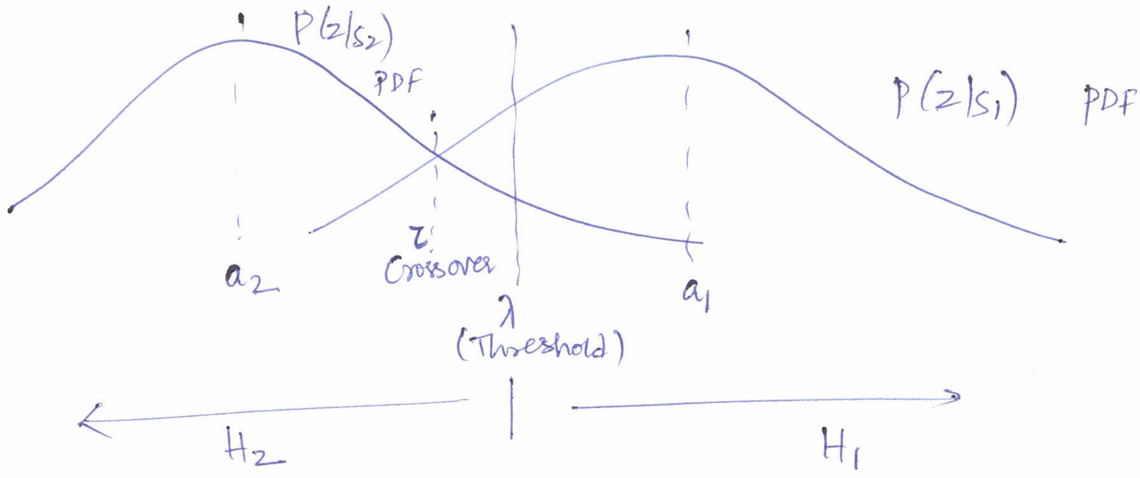
$$P_e = 1 - \sum_{i=1}^{M} p(s_i) \int_{z = s_i \text{ region}} p\left(z / s_i\right) dz$$

PDF

$\therefore$ MLI or MAP will reduce (lower) the P_e.

P_e for Two Symbol case

$p(s_1) \neq p(s_2)$



$$P_e = p(s_1)\, P(H_2|s_1) + p(s_2)\, P(H_1|s_2)$$

$$= p(s_1) \underbrace{\int_{-\infty}^{\lambda} p(z|s_1)\,dz}_{H_2 \text{ region}} + p(s_2) \underbrace{\int_{\lambda}^{\infty} p(z|s_2)\,dz}_{H_1 \text{ region}} \rightarrow (1)$$

$$= p(s_1) \int_{-\infty}^{\lambda} p(z|s_1)\,dz + p(s_2)\left[1 - \int_{-\infty}^{\lambda} p(z|s_2)\,dz\right]$$

$$= p(s_2) + \int_{-\infty}^{\lambda} \left[p(s_1)\, p(z|s_1) - p(s_2)\, p(z|s_2)\right] dz$$

To find the threshold 'λ' that minimizes the probability of error.
Set $\frac{dP_e}{d\lambda} = 0$ then solve and find λ. Use Leibniz's rule.

let $f(z,u) = p(s_1)\, p(z|s_1) - p(s_2)\, p(z|s_2)$

$a(u) = -\infty$
$b(u) = \lambda$

Leibniz's rule

$$\frac{d}{du} \int_{a(u)}^{b(u)} f(z,u)\,dz = f(b(u),u)\frac{db(u)}{du} - f(a(u),u)\frac{da(u)}{du} + \int_{a(u)}^{b(u)} \frac{\partial f(z,u)}{\partial u}\,dz = 0$$

$$\frac{d}{du}\int_{-\infty}^{\lambda} f(z,u)\,dz = f(\lambda,u)\frac{d\lambda}{du} - f(-\infty,u)\frac{d}{du}(-\infty) + \int_{-\infty}^{\lambda}\frac{\partial}{\partial u} f(z,u)\,dz = 0$$

$f(z,u)$ is independent of λ, 2nd and 3rd term of the above equation vanishes.

$$f(\lambda,u)\frac{\partial\lambda}{\partial u} = 0$$

$$\left[P(s_1)\,p(z=\lambda_0/s_1) - P(s_2)\,p(z=\lambda_0/s_2)\right] = 0$$

$$\frac{p(\lambda_0/s_1)}{p(\lambda_0/s_2)} = \frac{P(s_2)}{P(s_1)}$$

$$\frac{1}{\sqrt{2\pi\sigma^2}}\;\frac{e^{-(\lambda_0-a_1)^2/2\sigma^2}}{e^{-(\lambda_0-a_2)^2/2\sigma^2}} = \frac{P(s_2)}{P(s_1)} = \frac{e^{-(\lambda_0^2+a_1^2-2\lambda_0 a_1)/2\sigma^2}}{e^{-(\lambda_0^2+a_2^2-2\lambda_0 a_2)/2\sigma^2}}$$

$$e^{\left[(\lambda_0(a_1-a_2)/\sigma^2)-((a_1^2-a_2^2)/2\sigma^2)\right]} = \frac{P(s_2)}{P(s_1)}$$

$$\lambda_0\frac{a_1-a_2}{\sigma_0^2} - \frac{a_1^2-a_2^2}{2\sigma_0^2} = \ln\left[\frac{P(s_2)}{P(s_1)}\right]$$

$$\lambda_0 = \frac{a_1+a_2}{2} + \frac{\sigma^2}{a_1-a_2}\ln\frac{P(s_2)}{P(s_1)}$$ is the threshold for minimum P_e. → 2

When $P(s_1) = P(s_2) = \frac{1}{2}$, then $\lambda_0 = \frac{a_1+a_2}{2}$

from eqn ②, $\lambda = \frac{V_1+V_2}{2} + \frac{\sigma^2}{V_1-V_2} \ln \frac{P(S_2)}{P(S_1)}$

$$= \tau + \frac{\sigma^2}{V_1-V_2} \ln \frac{P(S_2)}{P(S_1)}$$

From eqn ①,

$$P_e = P(S_1)\int_{-\infty}^{\lambda} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(z-V_1)^2}{2\sigma^2}\right)dz + P(S_2)\int_{\lambda}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(\frac{-(z-V_2)^2}{2\sigma^2}\right)dz$$

let $\frac{z-V_1}{\sqrt{2}\sigma} = u$ $\qquad$ $\frac{z-V_2}{\sqrt{2}\sigma} = v$

$dz = \sqrt{2}\sigma\, du$ $\qquad$ $dz = \sqrt{2}\sigma\, dv$

$z \to -\infty,\ u \to -\infty$ $\qquad$ $z \to \infty,\ v \to \infty$

$z \to \lambda,\ u = \frac{\lambda - V_1}{\sqrt{2}\sigma} = \frac{-(V_1-\lambda)}{\sqrt{2}\sigma}$ $\qquad$ $z = \lambda,\ v = \frac{\lambda - V_2}{\sqrt{2}\sigma}$

$$P_e = P(S_1)\frac{1}{\sqrt{\pi}}\int_{-\infty}^{-\frac{V_1-\lambda}{\sqrt{2}\sigma}} \exp(-u^2)\,du + P(S_2)\frac{1}{\sqrt{\pi}}\int_{\frac{\lambda-V_2}{\sqrt{2}\sigma}}^{\infty} \exp(-v^2)\,dv$$

Gaussian function is symmetric.

$$P_e = P(S_1)\frac{1}{2}\left[\frac{2}{\sqrt{\pi}}\int_{\frac{V_1-\lambda}{\sqrt{2}\sigma}}^{\infty} \exp(-u^2)\,du\right] + P(S_2)\frac{1}{2}\left[\frac{2}{\sqrt{\pi}}\int_{\frac{\lambda-V_2}{\sqrt{2}\sigma}}^{\infty} \exp(-v^2)\,dv\right]$$

$$P_e = \frac{1}{2}P(S_1)\,\mathrm{erfc}\,\frac{V_1-\lambda}{\sqrt{2}\sigma} + \frac{1}{2}P(S_2)\,\mathrm{erfc}\left(\frac{\lambda-V_2}{\sqrt{2}\sigma}\right) \to ③$$

where λ is represented in eqn ②

Case (A) when $P(S_1) = P(S_2) \Rightarrow \lambda = \tau = \frac{V_1+V_2}{2}$

$$P_e = \frac{1}{4}\,\mathrm{Erfc}\left(\frac{V_1 - 0.5(V_1+V_2)}{\sqrt{2}\,\sigma}\right) + \frac{1}{4}\,\mathrm{Erfc}\left(\frac{0.5(V_1+V_2)-V_2}{\sqrt{2}\,\sigma}\right)$$

$$= \frac{1}{4}\,\mathrm{Erfc}\left(\frac{V_1-V_2}{2\sqrt{2}\,\sigma}\right) + \frac{1}{4}\,\mathrm{Erfc}\left(\frac{V_1-V_2}{2\sqrt{2}\,\sigma}\right)$$

$$P_e = \frac{1}{2}\,\mathrm{Erfc}\left(\frac{V_1-V_2}{2\sqrt{2}\,\sigma}\right) = \frac{1}{2}\,\mathrm{Erfc}\left(\frac{\Delta V}{2\sqrt{2}\,\sigma}\right)$$

as $\Delta V \uparrow$ Erfc $\downarrow$ $P_e \downarrow$ QoS $\uparrow$

<u>Case B</u>

$$P(S_1) = P\,,\quad P(S_2) = 1-P$$

$$\lambda = \tau + \frac{\sigma^2}{V_1-V_2}\ln\left(\frac{1-P}{P}\right)$$

$$P_e = \frac{P}{2}\,\mathrm{Erfc}\left(\frac{V_1-\lambda}{\sqrt{2}\,\sigma}\right) + \frac{1-P}{2}\,\mathrm{Erfc}\left(\frac{\lambda-V_2}{\sqrt{2}\,\sigma}\right)$$

Pe in terms of dmin

As long as distance b/w the symbols is maintained same, the Pe is Invariant to rotation and translation of a coordinate system.

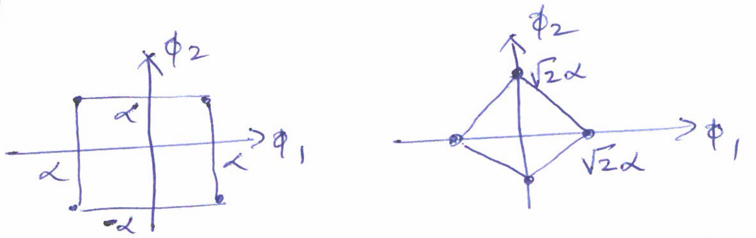


Fig - Rotation of the Constellation

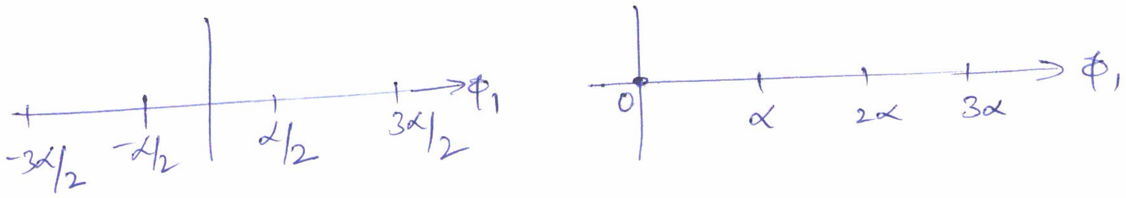
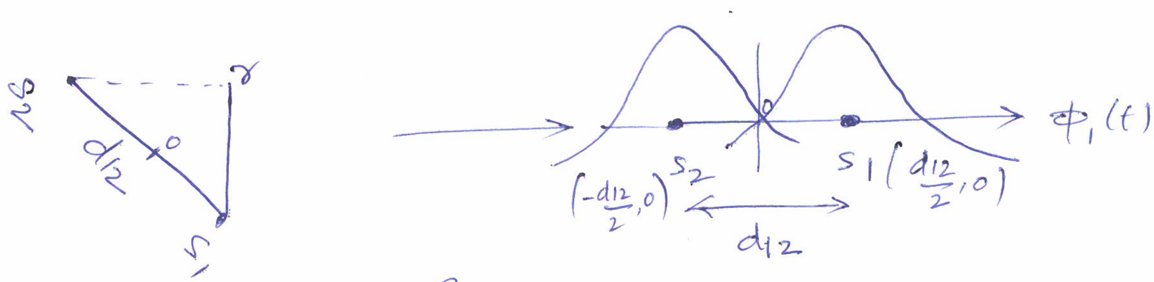


Fig - Translation of the Constellation.



$$P_e = \frac{1}{2}\left[\frac{2}{\sqrt{\pi}} \int_{\frac{d_{12}}{2\sqrt{2}\sigma}}^{\infty} \exp(-v^2)\, dv\right] = \frac{1}{2}\,\mathrm{Erfc}\left(\frac{d_{12}}{2\sqrt{N_0}}\right)$$

$$P_e = \frac{1}{2}\,\mathrm{Erfc}\left(\frac{d_{min}}{2\sqrt{N_0}}\right)$$

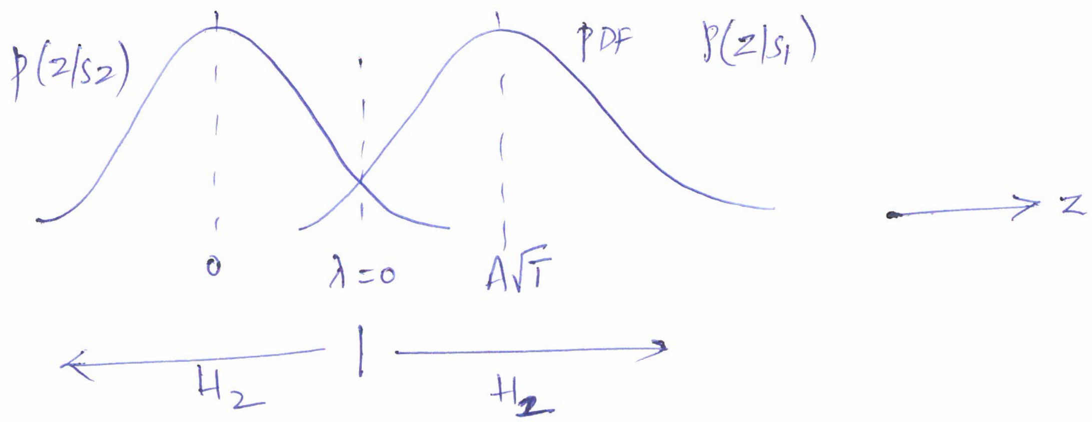
Procedure in page 4, Eqn (1) is followed to arrive at the above equation.

Probability of Error for Baseband Signals

$$P_e = \frac{1}{2}\,\mathrm{Erfc}\left(\frac{d_{min}}{2\sqrt{N_0}}\right) \quad , \quad \mathrm{Erfc}(x) = \frac{e^{-x^2}}{\sqrt{\pi}}$$

$$P_e \simeq \frac{1}{2\sqrt{\pi}}\exp\left(-\frac{d_{min}^2}{4N_0}\right) \qquad \text{Approximate Equation}$$

1. Unipolar Signal :



$$d_{min} = A\sqrt{T}$$

$$P_e = \frac{1}{2}\,\mathrm{Erfc}\left(\frac{A\sqrt{T}}{2\sqrt{N_0}}\right) = \frac{1}{2}\,\mathrm{Erfc}\left(\sqrt{\frac{E_{sav}}{2N_0}}\right)$$

$$E_{sav} = \frac{E_{s_1}+E_{s_2}}{2} = \frac{A^2T}{2}$$

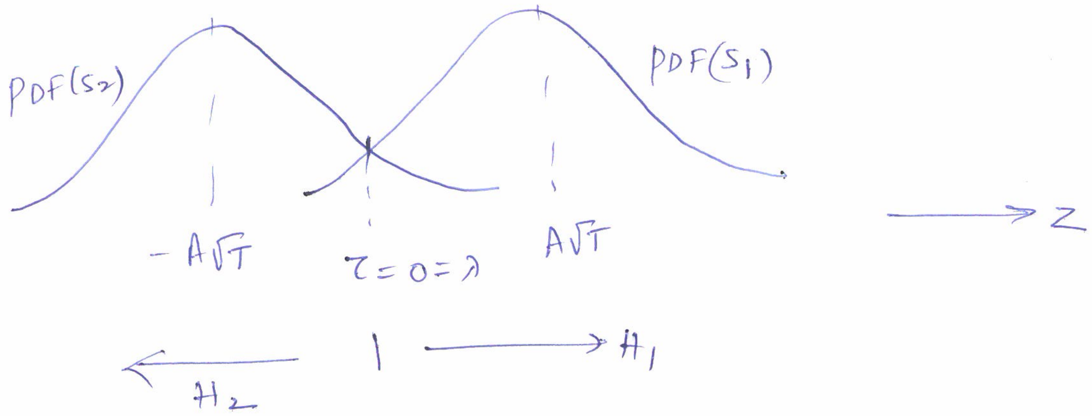
As $\frac{E_{sav}}{2N_0}\uparrow$ $P_e\downarrow$ QoS $\uparrow$

Conversions :

$$\mathrm{Erfc}(x) = 2\,Q(\sqrt{2}\,x) \quad , \quad Q(x) = \frac{1}{2}\,\mathrm{Erfc}\left(\frac{x}{\sqrt{2}}\right)$$

$$\mathrm{Erfc}(x) \simeq \frac{\exp(-x^2)}{\sqrt{\pi}} \quad , \quad Q(x) \simeq \frac{1}{\sqrt{2\pi}}\exp(-x^2/2)$$

2. polar signal



$$d_{min} = 2A\sqrt{T}$$

$$P_e = \frac{1}{2}\, \text{Erfc}\left(\frac{d_{min}}{2\sqrt{N_0}}\right) = \frac{1}{2}\, \text{Erfc}\left(\frac{A\sqrt{T}}{\sqrt{N_0}}\right)$$

$$= \frac{1}{2}\, \text{Erfc}\left(\sqrt{\frac{E_{sav}}{N_0}}\right)$$

$$E_{sav} = A^2 T$$

$\frac{E_{sav}}{N_0} \uparrow$ erfc $\downarrow$ $P_e \downarrow$ QoS $\uparrow$

$P_{e\,polar}$ is better than $P_{e\,unipolar}$.

Increase 3dB Energy in unipolar to achieve the P_e of polar.

3. Manchester Signal

P_e for manchester is same as P_e for polar signal.

Two symbols are antipodal in polar as well as manchester.

$$P_e = \frac{1}{2} \operatorname{Erfc}\left(\frac{A\sqrt{T}}{\sqrt{N_0}}\right) = \frac{1}{2} \operatorname{Erfc}\left(\sqrt{\frac{E_{sav}}{N_0}}\right)$$

4. Bipolar Signal :

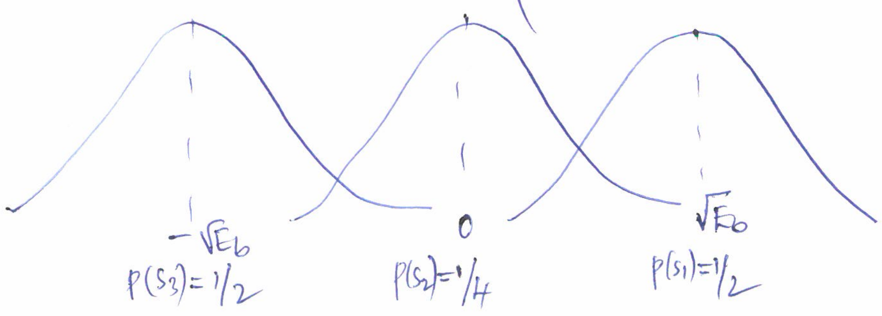
$$P_e = \underset{\substack{\uparrow \\ \text{Inner} \\ \text{symbols}}}{(M-2)} \; \underset{\substack{\uparrow \\ \text{Error both} \\ \text{sides}}}{\underbrace{2P_e}} \, P(0) + \underset{\substack{\uparrow \\ \text{outer} \\ \text{symbols}}}{2} \; \underset{\substack{\uparrow \\ \text{Error only one} \\ \text{direction}}}{P_e} \, P(+1)$$

for $M=3$ $\quad P_e = 1 \cdot 2P_e \frac{1}{2} + 2P_e \frac{1}{4} = P_e + \frac{1}{2}P_e = \frac{3}{2}\underset{\substack{\uparrow \\ P_e \text{ of} \\ \text{unipolar}}}{P_e}$

$$P_e = \frac{1}{2} \cdot \frac{3}{2} \operatorname{Erfc}\left(\sqrt{\frac{E_{sav}}{2N_0}}\right)$$

$$= \frac{3}{4} \operatorname{Erfc}\left(\sqrt{\frac{E_{sav}}{2N_0}}\right)$$

$$= \frac{3}{4} \operatorname{Erfc}\left(\sqrt{\frac{2A^2T}{3 \cdot 2N_0}}\right) = \frac{3}{4} \operatorname{Erfc}\left(\sqrt{\frac{A^2T}{3N_0}}\right)$$



P_e for Bandpass Signals

1. **BPSK**, $K=1, M=2$

$$P_e = \frac{1}{2} \, \mathrm{Erfc}\left(\frac{d_{min}}{2\sqrt{N_0}}\right)$$

$$d_{min} = 2\sqrt{E_b} = 2\sqrt{E_s}$$

$$P_e = \frac{1}{2} \, \mathrm{Erfc}\left(\frac{2\sqrt{E_b}}{2\sqrt{N_0}}\right) = \frac{1}{2} \, \mathrm{Erfc}\left(\sqrt{\frac{E_b}{N_0}}\right) = \frac{1}{2} \, \mathrm{Erfc}\left(\sqrt{\frac{E_s}{N_0}}\right)$$

2. **QPSK**, $k=2$, $M=4$

$$d_{min} = \sqrt{2E_s} \qquad E_s = 2E_b$$

$$P_e = \frac{1}{2} \, \mathrm{Erfc}\left(\sqrt{\frac{E_s}{2N_0}}\right) = \frac{1}{2} \, \mathrm{Erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

In terms of E_b, the P_e for BPSK and QPSK are same.

3. **MPSK**

$$d_{min} = 2\sqrt{E_s} \, \sin(\pi/M)$$

Optimistic P_e: In MPSK, Every symbol has two immediate neighbours. Consider only immediate neighbouring symbols for P_e calculations.

$$P_e = \frac{1}{M} \, 2 . (M-1)\left[\frac{1}{2} \mathrm{Erfc}\left(\frac{d_{min}}{2\sqrt{N_0}}\right)\right]$$

$$P_e \approx \operatorname{Erfc}\left(\frac{2\sqrt{E_s}\,\sin\pi/M}{2\sqrt{N_0}}\right) \approx \operatorname{Erfc}\sqrt{\frac{E_s}{N_0}\sin^2(\pi/M)}$$

$$E_s = kE_b = E_b\log_2 M$$

$$P_e \approx \operatorname{Erfc}\left[\sqrt{\frac{\sin^2(\pi/M)\,.\,E_b\log_2 M}{N_0}}\right]$$

Pessimestic P_e :- Consider all $(M-1)$ Symbols for calculating P_e for every symbol as adjacent symbols.

$$P_e = M\frac{1}{M}(M-1)\frac{1}{2}\operatorname{Erfc}\left(\frac{d_{min}}{2\sqrt{N_0}}\right)$$

$$= \frac{M-1}{2}\operatorname{Erfc}\left[\sqrt{\frac{\sin^2(\pi/M)\,E_b\log_2 M}{N_0}}\right]$$

4. M-PAM

Inner symbols $=(M-2)$

outer symbols $=2$

Optimestic P_e $$= \frac{1}{M}\left[(M-2)\,2.\frac{1}{2}\operatorname{Erfc}\left(\frac{d_{min}}{2\sqrt{N_0}}\right) + 2.1.\frac{1}{2}\operatorname{Erfc}\left(\frac{d_{min}}{2\sqrt{N_0}}\right)\right]$$

$$= \frac{M-1}{M}\operatorname{Erfc}\left(\frac{d_{min}}{2\sqrt{N_0}}\right)$$

Recall, d_{min} for PAM was $d_{min} = \sqrt{\frac{12\log_2 M}{M^2-1}\cdot E_{bavg}}$

$$P_e = \frac{M-1}{M}\operatorname{Erfc}\left(\sqrt{\frac{3\log_2 M}{M^2-1}\,\frac{E_{bavg}}{N_0}}\right)$$

pessimestic $Pe \simeq \frac{M}{M}(M-1)\frac{1}{2}\,\text{Erfc}\left(\sqrt{\frac{3\log_2 M}{(M^2-1)}\frac{E_{bavg}}{N_0}}\right)$

5. <u>M-QAM</u>

$$M\text{-QAM} = \sqrt{M}\ PAM \times \sqrt{M}\ PAM$$

<u>Optimestic Pe_{QAM}</u> $= 1 - Pc_{QAM}$

$$= 1 - \left[Pc_{\sqrt{M}PAM} \times Pc_{\sqrt{M}PAM}\right]$$

$$= 1 - \left[\left(1 - Pe_{\sqrt{M}PAM}\right)\left(1 - Pe_{\sqrt{M}PAM}\right)\right]$$

$$= 1 - \left[(1-x)(1-x)\right] = 1 - (1-x)^2 = 1 - (1 + x^2 - 2x)$$

$$\simeq 2x$$

$$Pe_{QAM} \simeq 2\,Pe_{\sqrt{M}PAM}$$

$$Pe_{QAM} = 2\left(1 - \frac{1}{\sqrt{M}}\right)\text{Erfc}\left(\sqrt{\frac{3\log_2\sqrt{M}}{(M-1)}\frac{E_{bavg}}{N_0}}\right)$$

<u>pessimestic Pe_{QAM}</u>

$$Pe \simeq (\sqrt{M}-1)\,\text{Erfc}\left(\sqrt{\frac{3\log_2\sqrt{M}}{(M-1)}\cdot\frac{E_{bavg}}{N_0}}\right)$$

6. <u>BFSK</u>

Coherent $Pe = \frac{1}{2}\,\text{erfc}\left(\sqrt{\frac{E_b}{2N_0}}\right)$

Non coherent $Pe \simeq \frac{1}{2}\exp\left(-\frac{E_b}{2N_0}\right)$

7. <u>DPSK</u>

$$Pe = \frac{1}{2}\exp\left(-\frac{E_b}{N_0}\right)$$